Depropanizer Column Ver 1 · OTS

DC3-V1-PE-RLB-001

Relief load basis

Rev 0 · Issued for teaching

Unit	Depropanizer Column Ver 1 · OTS
Document number	DC3-V1-PE-RLB-001
Title	Relief load basis
Discipline	Process (relief and flare)
Originator / DRAWN	Apinan Soottitantawat
Date	25-OCT-26
Source	simulator model (design snapshot and relief runs on copies of the model), epc/epc.json, tabs/erel/work/er.json

Revision record

REV	DATE	DESCRIPTION	BY	CHK	APP
0	25-OCT-26	Issued for teaching	A. Soottitantawat		

Note. *Numbers come from the simulator model (design case) and the assumptions listed; not for construction. Codes are named for the method only; no client, contractor, licensor or vendor is implied.*

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1 · Purpose and scope

Relief load basis of the C-201 depropanizer unit: for every pressure-relief device, each credible cause of overpressure (API 521), the relieving conditions, the relief load and the required orifice (API 520 Part I), with the inlet-loss and back-pressure checks (API 520 Part II), the flare / vent load summary and the depressuring note. Loads come from the design snapshot of the simulator model and the assumptions in section 2; the scenarios the model can represent were also run on copies of the model (section 5).

System protected by PSV-201 (set 1.80 MPa g, on the overhead line 8"-P-204): C-201, the E-202 shell and V-202, with no isolation valve between them. V-201 has no relief device on DC3-V1-PID-001 (note 11): PSV-202 is a recommended addition (EM-02). Thermal relief valves TSV-216, TSV-203 and TSV-202 are recommended additions.

Item	Design P MPa g	MAWP MPa g	Design T °C
C-201 top / bottom	1.90 / 1.950	2.001	140
E-202 shell	1.90	—	75
V-202	1.90	1.906	75
8"-P-204 overhead line	1.90	—	75
V-201 (PSV-202 recommended)	1.35	1.405	65

2 · Basis and assumptions

- Codes: API 521 (causes, relief loads, fire, thermal, tube rupture, depressuring), API 520 Part I (sizing, SI form) and Part II (inlet loss, back-pressure), API 526 (orifice areas), ASME VIII Div. 1 UG-125/134 (set ≤ MAWP).
- Set pressure 1.80 MPa g (model K.Psv, P&ID note 10); accumulation 10 % (single valve, 1.980 MPa g) and 21 % for fire (2.178 MPa g).
- Fluid properties from the model: pseudo-binary vapour pressures (dew and bubble points), molar latent heat λ = 15,500 kJ/kmol (model; real propane near 2 MPa is about 15–20 % lower, which would raise the fire loads), molar masses 44.1 / 58.1 kg/kmol (assumed, as the Design tab).
- Relief vapour: Z = 0.7 (assumed, as the Design tab); k = 1.13 y + 1.09 (1 − y) (assumed); K_d 0.975 (vapour), 0.65 (liquid), K_c = 1 (no rupture disc), K_b = 1 (conventional valve, critical flow).
- No credit for control action, operator action or the SIS in the loads (API 521); the model runs keep the trips and the PSV as in the model. No credit for the reduced heat input at relieving pressure in the FEED loads (the credited value is reported in section 4).
- Elevations (assumed, no plot plan yet): C-201 bottom tangent at +3.0 m (skirt), V-202 bottom of shell +1.5 m, E-202 shell +8.0 m (outside the 7.6 m fire zone), V-201 +1.0 m; drums wetted to the normal level 50 %.
- Fire: $Q = 43.2\ F\ A^{0.82}$ kW (adequate drainage and fire fighting), F = 1 (heat-conservation insulation gets no credit). Maximum LP-steam header pressure 0.70 MPa g (steam system design pressure, assumed).
- PSV-201 piping (assumed, no layout yet): inlet 4" with 3 m equivalent straight length and ΣK 0.7; tail pipe 6" (P&ID 4"×6") with 30 m equivalent length to a flare header at atmospheric pressure (design basis).

3 · Relief load basis

No.	Device	Scenario	Relieving P MPa g	Ph.	Load kg/h	kmol/h	M	T °C	Z	k	Area mm²	Ori.	Source
1	**PSV-201**	Total cooling-water failure (E-202)	1.980 (10 %)	V	45,205	945	47.8	77.7	0.70	1.119	2,022	**M**	calc + model
2	**PSV-201**	Reflux failure (both P-202 pumps lost)	1.980 (10 %)	V	45,205	945	47.8	77.7	0.70	1.119	2,022	**M**	calc (model for timing)
3	**PSV-201**	Reboiler steam valve TV-208 wide open at the maximum LP header pressure (0.70 MPa g)	1.980 (10 %)	V	11,644	263	44.3	53.2	0.70	1.129	520	**J**	calc + model
4	**PSV-201**	Blocked bottoms (LV-202 closed)	1.980 (10 %)	L	27.8 m³/h liq.	—	G 0.460	—	—	—	128	**F**	calc (model for timing)
5	**PSV-201**	General power failure	1.980 (10 %)	V	45,205	945	47.8	77.7	0.70	1.119	2,022	**M**	calc + model
6	**PSV-201**	External fire on C-201	2.178 (21 %)	V	8,819	153	57.7	123.8	0.70	1.091	352	**H**	calc
7	**PSV-201**	External fire on the reflux drum system (V-202; E-202 above the fire zone)	2.178 (21 %)	V	5,029	114	44.2	56.5	0.70	1.130	207	**G**	calc
8	**PSV-201**	Fire zone covering C-201 base and V-202 together	2.178 (21 %)	V	13,848	267	—	—	—	—	559	**J**	calc
9	**PSV-202 (recommended)**	External fire on V-201	1.633 (21 %)	V	6,810	146	46.6	60.9	0.70	1.123	361	**H**	calc
10	**PSV-202 (recommended)**	V-201 overfilled by the upstream unit (outlet blocked)	1.485 (10 %)	L	66.0 m³/h liq.	—	G 0.548	—	—	—	383	**H**	calc
11	**PSV-202 (recommended)**	V-201 blocked in, solar heating with a vapour space	—	—	—	—	—	—	—	—	no load	—	calc

Governing PSV-201 case: rows 1, 2 and 5 (no reflux): 45,205 kg/h, 2,022 mm² → orifice **M** (2323 mm², 4M6); the P&ID shows 3L4 / 4L6 (L, 1841 mm²) ✗ — query ER-01. PSV-202 (recommended) needs 383 mm² → H (1.5H3).

Basis of each row

No.	Scenario	Basis
1	Total cooling-water failure (E-202)	All overhead vapour relieved. V-202 empties (model run) and the reflux stops, so the vapour leaving the feed zone (tray 18 at design, light key 0.733) goes overhead; design heat input, no credit for the lower reboiler ΔT at relieving pressure (API 521 allows it, see part 3).
2	Reflux failure (both P-202 pumps lost)	The condenser keeps condensing until V-202 is full, then floods: no condensation and no reflux, so the relief load equals the no-reflux vapour of the cooling-water case. The model does not flood the condenser (the drum is unbounded) and its LSHH-203 trip stops feed and heat.
3	Reboiler steam valve TV-208 wide open at the maximum LP header pressure (0.70 MPa g)	Unbalanced heat: E-203 duty with TV-208 100 % open at 0.70 MPa g header and the bottoms at relieving pressure, minus the design duty, divided by the latent heat; reflux and condenser at their design duty (no credit for the condenser spare capacity).
4	Blocked bottoms (LV-202 closed)	Bottoms that cannot leave fill the column; the free volume above the normal sump level fills in 98 min. Credit for operator response to LAH-202 (normally taken for a fill time over 30 min); if not credited, liquid relief at the bottoms rate.
5	General power failure	Cooling-water pumps (assumed electric) and both reflux pumps stop: loss of condensing and of reflux together; feed continues on the steam-turbine spare P-201B; instrument air assumed held by its receiver. Same load as the cooling-water case.
6	External fire on C-201	API 521 $Q = 43.2\ F\ A^{0.82}$ kW (adequate drainage and fire fighting), F = 1 (heat-conservation insulation, no fire-proofing credit). Wetted area: bottom head and shell from BTL (+3 m) to 7.6 m above grade. Vapour from the boiling bottoms at 121 % of set.
7	External fire on the reflux drum system (V-202; E-202 above the fire zone)	V-202 wetted to the normal level 50 % (assumed elevation +1.5 m to the bottom of the shell); E-202 shell assumed at +8 m, outside the 7.6 m zone. Vapour = propane product boiling at 121 % of set.
8	Fire zone covering C-201 base and V-202 together	If C-201 and V-202 stand inside one fire area (assumed, ≤ 460 m²), both heat inputs reach PSV-201 at once: required area = sum of the two areas.
9	External fire on V-201	V-201 wetted to the normal level 50 % (bottom of shell +1 m, assumed); vapour from the feed boiling at 121 % of the 1.35 MPa g set. No PSV on V-201 in the OTS scope (EM-02): recommended addition.
10	V-201 overfilled by the upstream unit (outlet blocked)	Feed pumps stopped or FV-201 closed while the upstream unit keeps sending up to 110 % of the design rate; credible only if the upstream shut-off pressure exceeds 1.35 MPa g (not known: HOLD).
11	V-201 blocked in, solar heating with a vapour space	Feed bubble pressure at the 50 °C maximum metal temperature is 1.3 MPa g, below the 1.35 MPa g design (rule 2): no relief load while a vapour space remains.

Causes screened (API 521)

Cause	Outcome
Loss of cooling / loss of reflux / power failure	Rows 1, 2, 5 (governing)
Abnormal heat input (reboiler valve wide open; preheater valve wide open)	Row 3; the preheater case is smaller and covered
Blocked vapour outlet (PV-210 / HV-210 closed)	Load = normal vent 3.2 kmol/h, covered by row 1
Accumulation of non-condensables	Partial loss of condensing, bounded by row 1
Blocked bottoms; inadvertent feed at pump shut-off	Row 4 (time to fill 98 min, operator response credited)
Instrument-air failure	Fail positions open the vent and cut the heat: pressure falls, no load
Reverse flow	Debutanizer and fuel gas are below C-201: no load
Tube rupture; thermal expansion; external fire	Section 7; rows 6–9
Overfill / blocked-in V-201	Rows 10–11 (PSV-202 recommended)
Chemical reaction; vacuum	None; vacuum is EM-06

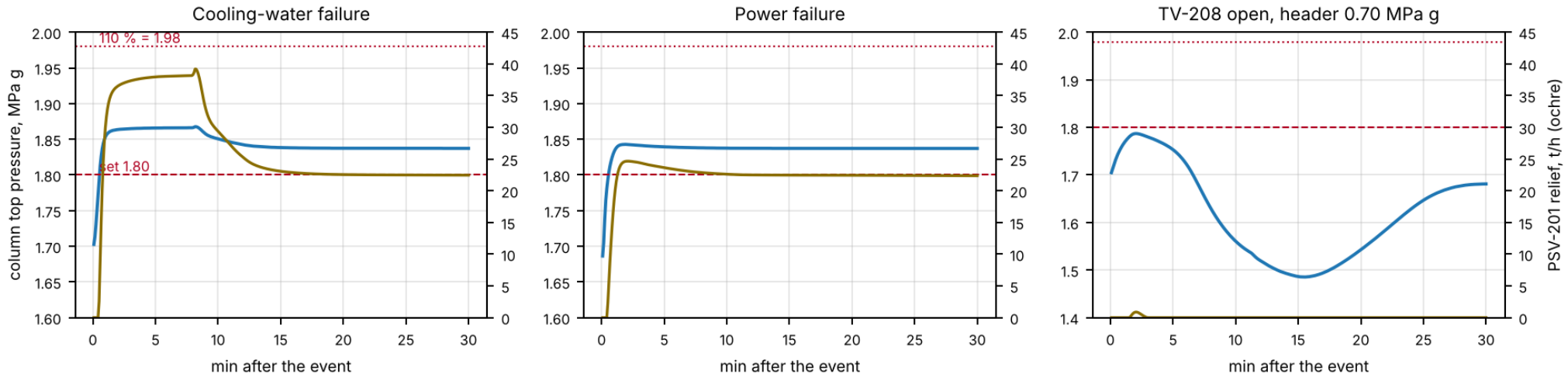
4 · Governing case, step by step

- Vapour reaching PSV-201: the reflux stops when V-202 empties (7:43 min:s after the cooling water is lost, model run), so the vapour leaving the feed zone (tray 18 at design) goes overhead: **945.0 kmol/h**, light key y = 0.733.
- M = 0.733 × 44.1 + 0.267 × 58.1 = 47.84 kg/kmol; W = **45,205 kg/h**.
- P_1 = 1.10 × 1,800 + 101.3 = 2,081.3 kPa a; T = dew point of y at P_1 = **77.7 °C**; Z 0.7, k 1.119.
- Critical flow (101.3 / 2,081 = 0.049 below r_c); C = 0.02497.
- A = W / (C K_d P_1) √(T Z / M) = 45,205 / (0.02497 × 0.975 × 2,081.3) × √(350.9 × 0.7 / 47.84) = **2,022 mm²** → orifice **M** (2323 mm², 4M6).
- Design tab basis (reflux running): 920 kmol/h, y 0.985, 40,767 kg/h at 53.2 °C → 1,821 mm² (L).
- With API 521 credit for the lower heat input at relieving pressure (bottoms boil at 117.9 °C, E-203 1,923 kW instead of 2,239 kW; feed flashes 26 % instead of 67 %): 611 kmol/h → 1,471 mm² (L). FEED keeps the conservative value (ER-01).

5 · Model relief runs

Fresh copy of the design snapshot for each run (noise off, step 0.25 s), scenario applied at t = 0, no operator action; controllers, trips LSLL-202 / LSHH-203, auto-starts and PSV-201 as in the model (W_{psv} = K_{psv} ((P_{top} − (P_{set} − 20))/100)$^{1.5}$ kmol/s, K_{psv} 0.3, full lift 1.91 MPa g, 1,601 kmol/h). Script tabs/erel/work/mk_data.js; the page re-runs any case.

Run	First opens	Set reached	Peak P MPa g	Peak relief kg/h	To flare	SIS trip	Other marks	RLB load
Cooling-water failure	0:27	0:32	1.868 at 8:12	39,197	6,476 Nm³	none	V-202 < 5 % at 7:43	45,205
Reflux failure	—	—	1.701 at 0:09	0	0	LSHH-203 at 5:07	V-202 92 % at 5:05	45,205
TV-208 open, header 0.70 MPa g	1:30	—	1.787 at 2:03	877	6 Nm³	none	—	11,644
TV-208 open, header 0.45 MPa g	—	—	1.736 at 3:12	0	0	none	—	—
Blocked bottoms	—	—	1.700 at 27:55	0	0	none	sump LAH at 4:25; sump 100 % at 8:35	27.8 m³/h liq.
Power failure	0:25	0:33	1.843 at 1:51	24,666	5,199 Nm³	none	—	45,205



The model does not represent fire, tube rupture, a condenser flooded by a full V-202, liquid filling of the column or the V-201 pressure: those rows are hand calculations. Sensitivity to the PSV relation (ER-03):

Run	PSV as in the model	scaled to L (×0.563)	scaled to M (×0.710)	PSV not credited
Cooling-water failure, peak MPa g	1.868	1.907	1.889	3.291
Power failure, peak MPa g	1.843	1.870	1.858	—

6 · PSV-201 sizing, inlet loss and back-pressure

Quantity	Value	Criterion / note
Governing required area	2,022 mm²	rows 1, 2, 5
Selected orifice	M (2323 mm², 4M6)	P&ID L (1841 mm²) ✗ ER-01
Rated capacity of M at the governing conditions	51,939 kg/h	inlet and outlet checked at this flow
Inlet line 4": velocity / loss	36.0 m/s / 37.3 kPa = 2.07 % of set	✓ ≤ 3 %
Built-up back-pressure, 6" tail pipe	353 kPa = 19.6 % of set (exit choked)	✗ ≤ 10 % conventional — ER-02
Built-up back-pressure, 8" tail pipe	137 kPa = 7.6 % of set	✓ ≤ 10 %
PSV-202 (recommended, V-201)	383 mm² → H (1.5H3)	overfill liquid 383 mm², fire 361 mm² — ER-05

7 · Fire, thermal relief and tube rupture

Fire case	Wetted area m²	Q kW	λ kJ/kg	Load kg/h	T °C	Area mm²	Orifice
External fire on C-201	27.70	658	268.6	8,819	123.8	352	H
External fire on the reflux drum system (V-202; E-202 above the fire zone)	19.35	490	351.0	5,029	56.5	207	G
External fire on V-201	26.22	629	332.6	6,810	60.9	361	H

Device	Blocked-in liquid	φ kW	Q m³/h	Set MPa g	Area mm²	Orifice
TSV-216 (rec.)	E-202 cooling-water side (tubes), blocked in by HV-216 and the return block valve	5,882.1	1.53	0.70	16.6	D
TSV-203 (rec.)	Propane rundown 3"-P-213 between LV-203 and the storage block valve	4.0	0.040	2.60	0.2	D
TSV-202 (rec.)	E-201 tube side (LPG feed) blocked in between FV-201 / XV-201 and the column with LP steam still on the shell	7,057.7	55.59	2.85	222.1	G

Exchanger	High side design / max op MPa g	Low side design / max op MPa g	10/13 × high	Credible	Rate kg/h	Query
E-202	Shell (overhead vapour) 1.90 / 1.702	Tubes (cooling water) 0.70 / 0.450	1.46	yes ✗	5,382, vapour	EM-03
E-201	Tubes (LPG feed) 2.85 / 2.563	Shell (LP steam) 0.70 / 0.260	2.19	yes ✗	42,304, liquid (flashes)	EM-04
E-203	Tubes (C-201 bottoms) 1.95 / 1.740	Shell (LP steam) 0.70 / 0.304	1.50	yes ✗	30,581, liquid (flashes)	EM-04

Tube ¾" OD × 16 BWG (ID 15.75 mm, assumed), two open ends; vapour C_d 0.7, liquid C_d 0.6 without flashing (conservative). Thermal rate $Q = \alpha \varphi / (\rho c_p)$.

8 · Flare and vent load summary

Event	PSV-201 kg/h	PSV-202 (rec.) kg/h	Total to flare kg/h	PV-210 to fuel gas, model peak Nm³/h
Cooling-water failure (site-wide)	45,205	—	**45,205**	538
Reflux failure	45,205	—	**45,205**	74
General power failure (site-wide)	45,205	—	**45,205**	533
TV-208 wide open, header 0.70 MPa g	11,644	—	**11,644**	209
Fire on C-201	8,819	—	**8,819**	—
Fire on V-202	5,029	—	**5,029**	—
Fire zone C-201 + V-202	13,848	—	**13,848**	—
Fire on V-201	—	6,810	**6,810**	—
Depressuring under fire (BDV-201, if added)	—	—	48,217 (initial, not a PSV load)	—

Governing flare load of this unit 45,205 kg/h (M 47.8, 78 °C). Cooling-water and power failure are site-wide: the flare header and knock-out drum (outside this unit) add the other units' loads for the same event.

9 · Depressuring note (recommendation only)

API 521 guidance: vapour depressuring is normally considered for equipment operating at about 1.7 MPa g (250 psig) and above and for LPG equipment exposed to fire; the usual target is 690 kPa g or 50 % of the design pressure, whichever is lower, in about 15 min for walls of 25 mm and more (thinner walls need faster depressuring). C-201 (18 mm shell, design 1.90 MPa g) operates at 1.70 MPa g: target 0.69 MPa g in 15 min.

Inventory	kmol	kg	T °C	Bubble point at 0.69 MPa g	Flash kmol
C-201 sump	38.7	2,245	109.0	64.1	14.0
C-201 trays 1–27	42.9	2,194	70.4	27.6	14.8
V-202 reflux drum	89.0	3,941	39.4	8.6	22.1
Vapour (model gas volume 120 m³)	—	4,921	—	—	68.7
Fire heat for 15 min	—	—	—	—	66.7

Average 745 kmol/h; initial ≈ 1.46 × average = 1,088 kmol/h ≈ 48,217 kg/h. Flash cooling of the V-202 propane: 8.6 °C at 0.69 MPa g, −43.6 °C at 50 kPa g (MDMT −46 °C, EM-01).

Recommendation. Provide BDV-201 (recommended addition; fail-open, to the flare) on the V-202 vapour space, which depressures C-201 through E-202; size it with a dynamic depressuring study in detailed engineering. If not added, record the justification (fire-resistant insulation of the C-201 skirt and V-202, water deluge, fire-risk assessment) — ER-07.

10 · Checks against the model and the other documents

Check	Value	Against	Result	Query
PSV-201 set ≤ MAWP of C-201 / V-202	1.80 MPa g	2.001 / 1.906	✓	
Design pressure of C-201, E-202 shell, V-202, 8"-P-204 ≥ set	1.80 MPa g	1.90 MPa g	✓	
Set ≥ 110 % of operating	1.80	1.870	✗	EP-01
Governing area ≤ P&ID orifice L	2,022 mm²	1841 mm²	✗	ER-01
Governing area ≤ selected orifice M	2,022 mm²	2323 mm²	✓	
Model peak relief ≤ RLB load (cooling-water failure)	39,197	45,205	✓	
Model peak relief ≤ RLB load (power failure)	24,666	45,205	✓	
Model peak pressure ≤ 110 % of set (cooling water / power / TV-208)	1.868 / 1.843 / 1.787	1.980	✓	
Model PSV capacity ≈ rated M (± 25 %)	1,601 kmol/h	1,137 kmol/h	✗	ER-03
Inlet loss ≤ 3 % of set	2.07 %	3 %	✓	
Built-up back-pressure ≤ 10 % (6" tail pipe)	19.6 %	10 %	✗	ER-02
Relieving T ≤ design T of V-202 / E-202 shell / 8"-P-204 (non-fire)	77.7 °C	75 °C	✗	ER-04
Fire relieving T ≤ design T of 8"-P-204	123.8 °C	75 °C	✗	ER-04
PIC-210 / PI-211 high alarms below set	1.76 / 1.78 MPa g	1.80	✓	
V-201 has a PSV	none	PSV-202 recommended	✗	EM-02, ER-05
V-201 design ≥ feed bubble pressure at 50 °C	1.35	1.301	✓	
E-202 low side ≥ 10/13 of high side	0.70	1.46	✗	EM-03
E-201 low side ≥ 10/13 of high side	0.70	2.19	✗	EM-04
E-203 low side ≥ 10/13 of high side	0.70	1.50	✗	EM-04
Blocked bottoms: time to fill ≥ 30 min	98 min	30 min	✓	
Depressuring to 50 kPa g: lowest T ≥ MDMT	−43.6 °C	−46 °C	✓	EM-01

11 ✓, 10 ✗ (the page recomputes the same checks live, with a few more).

11 · Engineering queries raised

No.	Subject	Query	Proposed answer	Status
ER-01	PSV-201 orifice: L on the P&ID, M required	The governing PSV-201 cases (cooling-water failure, reflux failure and power failure, all ending without reflux) need 2,022 mm² for 45,205 kg/h at 1.98 MPa g, more than the L orifice of DC3-V1-PID-001 note 10 (1841 mm²). The Design tab sized L for the overhead vapour with reflux still running (40,767 kg/h, 1,821 mm²); the model run shows V-202 empties 7.7 min after the cooling water is lost, after which the reflux stops.	Specify PSV-201 as 4M6 (M, 2323 mm²) in the P&ID and the PSV datasheet; or keep L only if a dynamic relief study credits the lower heat input at relieving pressure (611 kmol/h, 1,471 mm², orifice L).	OPEN
ER-02	PSV-201 built-up back-pressure with a 6" tail pipe	At the rated capacity of the M orifice (51,939 kg/h) a 6" tail pipe of 30 m equivalent length (assumed) to an atmospheric flare header builds up 353 kPa g = 19.6 % of set (choked at the exit): above the 10 % limit of a conventional valve.	Enlarge the tail pipe to 8" right at the valve outlet (7.6 % of set) or specify a balanced-bellows valve; confirm the superimposed back-pressure of the flare header (the basis assumes atmospheric) and add it to the check.	OPEN
ER-03	Model PSV-201 relation larger than the selected orifice	The simulator relation Wpsv = K.Kpsv × ((P − (set − 20))/100)$^{1.5}$ gives 1,601 kmol/h at full lift (1.91 MPa g), 1.78 × the API 520 rated capacity of an L orifice and 1.41 × an M orifice. Model peak pressures are therefore slightly optimistic: with K.Kpsv scaled to the L orifice the cooling-water-failure peak is 1.907 MPa g (model 1.868).	In the next model revision set K.Kpsv = 0.213 (M orifice) so that the simulated PSV flows match the relief load basis; no change to the documents.	OPEN
ER-04	Relieving temperatures above the overhead-system design temperature	The no-reflux cases (cooling water, reflux, power failure) relieve the vapour of the feed zone at its dew point 77.7 °C at 1.98 MPa g, above the 75 °C design temperature of 8"-P-204, the E-202 shell and V-202 (rule (a): 45 °C + 28 °C). A fire on C-201 relieves vapour from the boiling bottoms at 124 °C (121 % of set) through 8"-P-204 and the PSV-201 inlet (C-201 itself is designed for 140 °C).	Raise the design temperature of 8"-P-204, the PSV-201 inlet line and body, the E-202 shell and V-202 to 85 °C (relieving temperature + margin); record the fire case as an occasional variation accepted under ASME B31.3 302.2.4 or design the PSV inlet line for 140 °C; the PSV datasheet states both relieving temperatures.	OPEN
ER-05	PSV-202 on V-201: liquid relief and upstream pressure	The recommended PSV-202 (set 1.35 MPa g) is sized by the overfill case: 66 m³/h of liquid LPG (383 mm²) against fire 6,810 kg/h (361 mm²); both fit an H orifice. The overfill case exists only if the upstream shut-off pressure exceeds 1.35 MPa g (EM-02).	Confirm the upstream maximum pressure; if overfill is credible route PSV-202 to the flare through a knock-out drum able to take liquid, otherwise size PSV-202 for fire only.	OPEN
ER-06	Thermal relief of the E-201 tube side	LPG blocked in the E-201 tubes with steam on the shell expands at 55.6 m³/h at the start (UA × ΔT = 7,058 kW): a G orifice, not a ¾" × 1" thermal valve, and the liquid soon reaches its bubble point.	Add TSV-202 on the E-201 tube outlet (recommended addition) relieving to a closed system, and close XV-202 whenever the feed is isolated (the SIS feed trip already closes XV-201 and XV-202 together).	OPEN
ER-07	Depressuring valve for C-201 / V-202	C-201 operates at 1.70 MPa g with about 13.3 t of hydrocarbon in C-201 and V-202. Depressuring to 0.69 MPa g in 15 min under fire needs about 33,025 kg/h on average and 48,217 kg/h at the start, above the PSV-201 governing load.	Recommended: BDV-201 (fail-open, to flare) on the V-202 vapour space, sized in detailed engineering by a dynamic depressuring study; the flare header then takes the depressuring load. Alternative: fire-proofing and water deluge of C-201 skirt and V-202 with a documented risk assessment.	OPEN

Earlier queries this document answers or depends on

No.	Subject	Relief finding
EP-01	PSV-201 set pressure vs operating pressure	Still open (set 5.9 % above operating); the orifice choice of ER-01 does not change with a 1.90 MPa g set.
EM-01	Minimum design metal temperature and material	Depressuring to 50 kPa g cools the V-202 propane to −43.6 °C: MDMT −46 °C holds.
EM-02	V-201 design pressure and overpressure protection	PSV-202 recommended, set 1.35 MPa g, 1.5H3 (rows 9–11, ER-05).
EM-03	E-202 tube rupture (cooling-water side)	Tube rupture 5,382 kg/h of propane vapour into the cooling water.
EM-04	E-201 / E-203 steam side vs process side	Tube rupture 42,304 / 30,581 kg/h of liquid hydrocarbon into the E-201 / E-203 steam shells.